

ATLIQ MEDIA · CODEBASICS RPC #21

Tamil Nadu Assembly Election 2026

AtliQ Media — Neutral Election Briefing

PROJECT OVERVIEW

ECI public data only · 234 constituencies · 2021 vs 2026

Generated: May 28, 2026 · Descriptive analysis · No predictions, no causal claims

1 · PROJECT OVERVIEW

Project Overview

Tamil Nadu Assembly 2026 — AtliQ Election Briefing

Interactive briefing dashboard for **AtliQ Media**: neutral analysis of Tamil Nadu Legislative Assembly results (2021 vs 2026), built from public Election Commission of India data only.

Codebasics Resume Project Challenge #21.

Disclaimer

Non-partisan data work. No political endorsement, predictions, or causal claims about why parties gained or lost seats.

Quick start

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
python scripts/build_processed_data.py

# API
uvicorn api.main:app --reload --port 8000

# Dashboard (separate terminal)
cd web && npm install && npm run dev
```

Open <http://localhost:5173>. The UI proxies `/api` to port 8000.

Production build (single port):

```
cd web && npm run build
uvicorn api.main:app --host 127.0.0.1 --port 8000
```

Share externally: `bash scripts/ngrok_share.sh`

What you see in the app

Section	Purpose
Editorial brief	Recommended 60-minute run-of-show and three core narrative threads
Margins	Fragmentation — margins, vote-share buckets, closest races
Seat flows	2021→2026 Sankey and incumbent retention
Statewide	KPIs, mosaic, seat tally, vote share
Explorer	Filters + sortable constituency table
Geography / Reserved / Ballots	Supporting views — regional vote share, reserved comparison, turnout
Deep insights	Electoral arithmetic — ENP, Pedersen volatility, Gallagher LSq, swing heatmap, representation gap, district churn
Sources	Definitions, data scope, caveats

Insight bullets on each page are computed from the same `data/processed/` tables as the charts.

Data & metrics

File	Role
<code>data/raw/tn_2021_results.csv</code>	2021 candidate-level results
<code>data/raw/tn_2026_results.csv</code>	2026 candidate-level results
<code>data/raw/constituency_master.csv</code>	234 ACs — district, macro-region, reserved
<code>data/processed/</code>	Joined AC tables, Sankey edges, regional aggregates

Join on `ac_number` (not constituency name). Definitions: `docs/METRIC_DEFINITIONS.md`.

Note: 2026 `turnout` is blank in the provided CSV. The pipeline reads `data/external/turnout_2026.csv` if present (template via `python scripts/fetch_eci_turnout_2026.py --template-only`); once filled from ECI Form-20, the `/depth` page automatically shows the Top-20 turnout-increase chart.

Headline numbers (this build)

- **108** seats — normalized winner TVK (2026)
- **163** normalized seat flips (2021→2026), ~70% of ACs
- Mean winning margin **11.8%** (2021) → **7.7%** (2026)

- **61** winners below **35%** of valid votes (2026)

Deep electoral arithmetic (Deep Insights page)

- **Effective Number of Parties** (Laakso–Taagepera): **3.74 → 4.29** statewide
- **Pedersen volatility**: **36.4** — academic literature flags 20+ as "very high"
- **Gallagher LSq disproportionality**: **15.2 → 8.7** — seats track votes more closely in 2026
- **132 of 234** ACs were multi-cornered races in 2026 (top-2 combined share < 70%)
- Definitions and constitutional context in [docs/METRIC_DEFINITIONS.md](#)

Deck, video, and exports

```
python scripts/build_all_deliverables.py      # processed data + deck + charts +  
walkthrough  
# Or step by step:  
python scripts/build_processed_data.py  
python scripts/build_presentation.py         # deck/TN_2026_AtliQ_Briefing.pptx (18 slides)  
python scripts/export_charts.py              # outputs/charts/  
python scripts/generate_walkthrough.py       # outputs/walkthrough/
```

Narration cues: [docs/VIDEO_NARRATION_SCRIPT.md](#). QA: [docs/DASHBOARD_QA.md](#). Reuse another election: [docs/DASHBOARD_REUSE.md](#).

Tests

```
cd web && npm run test:e2e  
python scripts/test_dashboard_api.py        # API on :8000
```

Repository map

Path	Purpose
<code>web/</code>	React + TypeScript + ECharts UI
<code>api/</code>	FastAPI over processed CSVs
<code>src/</code>	Metrics, normalization, charts
<code>scripts/</code>	Data build, exports, ngrok helper
<code>notebooks/</code>	Reproducible analysis
<code>config/election.json</code>	Jurisdiction metadata
<code>HACKATHON_EXECUTION_PLAN.md</code>	Internal execution checklist
<code>docs/DASHBOARD_QA.md</code>	Manual + E2E QA
<code>docs/DASHBOARD_REUSE.md</code>	Reuse pipeline for another election
<code>docs/VIDEO_NARRATION_SCRIPT.md</code>	Recording script aligned to routes
<code>deck/</code>	PowerPoint for video + RPC deck deliverable

Reproduce from a clean clone

```
rm -rf data/processed/*
python scripts/build_processed_data.py
uvicorn api.main:app --port 8000
```